2025 11th International Workshop on Signal Design and Its Applications in Communications (IWSDA) | 979-8-3315-9340-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/IWSDA67899.2025.11457425

Construction of Codes Over $\mathbb{F}_{p^3}$ From Cyclic Codes Over $\mathscr{M}_3(\mathbb{F}_p + u\mathbb{F}_p + u^2\mathbb{F}_p)$

Shikha Patel[1], Soham Ravikant Joshi[2] and Om Prakash[3]
[1]Department of Mathematics, Indian Institute of Information Technology Bhopal, Bhopal- 462 003, India
[2,3]Department of Mathematics, Indian Institute of Technology Patna, Bihta, Patna - 801 106, India
Email: shikha_1821ma05@iitp.ac.in, soham_2421ma14@iitp.ac.in, om@iitp.ac.in

***Abstract*—Let $\mathbb{F}_p$ be a finite field with p elements and $\mathbb{F}_p[u]/\langle u^3\rangle = \mathbb{F}_p + u\mathbb{F}_p + u^2\mathbb{F}_p,\ u^3 = 0$ be a finite chain ring. This paper presents the construction of cyclic codes over $\mathbb{F}_{p^3}$ for certain primes $(p \leq 100)$ from a noncommutative non-chain ring $\mathscr{R} = \mathscr{M}_3(\mathbb{F}_p[u]/\langle u^3\rangle)$ of 3×3 matrices. First, we show that the structure of $\mathscr{R}$ is isomorphic to $(\mathbb{F}_{p^3}+e\mathbb{F}_{p^3}+e^2\mathbb{F}_{p^3}+u(\mathbb{F}_{p^3}+e\mathbb{F}_{p^3}+e^2\mathbb{F}_{p^3})+u^2(\mathbb{F}_{p^3}+e\mathbb{F}_{p^3}+e^2\mathbb{F}_{p^3})$ where $e^3 = 1, u^3 = 0, u^ie^j = e^ju^i$ for $i \in \{1,2\}$ and $j \in \{1,2,3\}$. Then we provide the form of ideals of $\mathscr{R}$ and derive that cyclic codes can be written as the direct sum of $\mathscr{R}$-submodules of $\frac{\mathscr{R}[x]}{<x^n-1>}$. Further, we find the formula for the cardinality of cyclic codes over $\mathscr{R}$. Later, in the module isometry for $\mathscr{R}$, we define the Gray map, which takes derived cyclic codes over $\mathscr{R}$ to $\mathbb{F}_{p^3}$. In addition, we provide non-trivial examples of linear codes over $\mathbb{F}_{p^3}$ with good parameters which support our derived results.*

***Index Terms*—Finite field, cyclic codes, matrix rings, Gray map, MDS/AMDS codes.**

I. Introduction

The study of cyclic codes over finite rings has received considerable attention in the last decades of the twentieth century and has made significant contributions to the development of error-correcting codes. Several studies have been conducted over the past three decades on cyclic codes over finite fields and finite commutative rings. Many well-known codes, including binary Hamming codes, Reed-Muller, Golay, Kerdock, BCH, Preparata, and Justesen codes, are either cyclic or constructed from cyclic codes. Recall that a cyclic code of length n over a finite field $\mathbb{F}_q$ is also classified as the ideal of the quotient ring $\mathbb{F}_q[x]/\langle x^n - 1\rangle$. Due to the rich algebraic structure, researchers prefer the class of cyclic codes for encoding and decoding purposes over conventional linear codes.

To obtain a new code with better parameters, cyclic codes over finite commutative rings were investigated in [13], [15], [16], [18]. Matrix rings also produce distinct and good cyclic codes, much like skew polynomial rings. This was accomplished by Alahmadi et al. [1], who obtained different self-dual $\mathbb{F}_4$-codes and cyclic codes structure over $M_2(\mathbb{F}_2)$ concerning Euclidean self-dual cyclic values. In 2014, Falcunit and Sison [5] further investigated the algebraic properties of cyclic codes over a more extended ring $M_2(\mathbb{F}_p)$, where $p \equiv 2$ or $3 \pmod 5$. Furthermore, Luo and Udaya [9] investigated cyclic codes over the matrix ring $M_2(\mathbb{F}_2[u]/\langle u^2\rangle)$ in 2018. Several quaternary optimal codes were obtained by using an isometry between $M_2(\mathbb{F}_2+u\mathbb{F}_2)$ and $\mathbb{F}_4[u,v]/\langle u^2, v^2, uv-vu\rangle$. Later, Bhowmick et al. [2] studied self-dual cyclic codes over $M_2(\mathbb{Z}_4)$, and Pal et al. [12] studied cyclic codes over $M_4(\mathbb{F}_2)$. The cyclic codes over $M_2(\mathbb{F}_p + u\mathbb{F}_p)$ were also examined by Islam et al. [7] in 2021. Further, in 2022, Patel et al. [14] discussed cyclic codes over $M_4(\mathbb{F}_2 + u\mathbb{F}_2)$ and obtained many good codes over $\mathbb{F}_{16}$. More recently, in [8], the authors considered the cyclic codes over $\mathscr{M}_3(\mathbb{F}_p)$ in 2024 and discovered good linear codes over $\mathbb{F}_8$ for $p = 2$. Continuing with that framework, we find linear codes over different fields with better parameters. Two significant contributions of the paper are given below.

- We analyze and explore the algebraic structure of the noncommutative non-chain ring $\mathscr{M}_3(\mathbb{F}_p + u\mathbb{F}_p + u^2\mathbb{F}_p)$ with $u^3 = 0$.
- We obtain some linear codes which are either MDS (maximum-distance-separable) or AMDS (almost maximum-distance-separable).

The paper is organized as follows: Section II establishes the structure of $\mathscr{R} = \mathscr{M}_3(\mathbb{F}_p + u\mathbb{F}_p + u^2\mathbb{F}_p)$ while Section III studies the cyclic code on $\mathscr{R}$ as the direct sum of the submodules of $\mathscr{R}$. Section IV defines the Gray map for the correspondence between $\mathbb{F}_{p^3}^9$ and $\mathscr{R}$, and Section V includes some examples of non-trivial codes to support our obtained results. Finally, Section VI concludes the paper.

II. Construction of $\mathscr{M}_3(\mathbb{F}_p + u\mathbb{F}_p + u^2\mathbb{F}_p)$

Let $\mathbb{F}_p$ be the finite field of p elements. Throughout the paper, $\mathscr{R}$ represents the ring $\mathscr{M}_3(\mathbb{F}_p[u]/\langle u^3\rangle)$, which is a finite noncommutatie non-chain ring. For a prime $p \leq 100$, let $f(x) \in \mathbb{F}_p[x]$, an irreducible polynomial of degree 3. For example, $f(x) = x^3 - x - 1$ is irreducible in $\mathbb{F}_p[x]$ for $p \in \{2,3,13,29,31,41,47,71,73\}$ as it has no roots in $\mathbb{F}_p$. Similarly, $g(x) = x^3 + x + 1$ is irreducible in $\mathbb{F}_p[x]$ for $p \in \{2,5,7,19,41,59,71,97\}$ and $h(x) = x^3 + 3x + 2$ is irreducible in $\mathbb{F}_p[x]$ for $p \in \{5,7,11,31,53,73,83,97\}$. We can choose any of $f(x)$, $g(x)$, or $h(x)$ depending on which prime we have to construct $\mathbb{F}_{p^3} = \mathbb{F}_p[\omega]$ which is a Galois field where $f(\omega) = 0$ or $g(\omega) = 0$ or $h(\omega) = 0$. The Galois group $G = Gal(\mathbb{F}_{p^3}/\mathbb{F}_p)$ is cyclic generated by Frobenius automorphism $\sigma : \mathbb{F}_{p^3} \longrightarrow \mathbb{F}_{p^3}$ defined by $\sigma(a) = a^p$. Also, using the construction given in [8], [11], we have a cyclic algebra $\mathbb{A} = (\mathbb{F}_{p^3}/\mathbb{F}_p, \sigma, 1) = \mathbb{F}_{p^3} + e\mathbb{F}_{p^3} + e^2\mathbb{F}_{p^3}$ where $ae = e\sigma(a)$, and $e^3 = 1$ for all $a \in \mathbb{F}_{p^3}$. Here, $\mathbb{A}$ is a central

Authorized licensed use limited to: SR University. Downloaded on November 12,2026 at 06:28:05 UTC from IEEE Xplore. Restrictions apply.

simple algebra with $dim(\mathbb{A}) = 9$ and therefore, $\mathbb{A} \cong \mathscr{M}_3(\mathbb{F}_p)$. It is easy to check that

$$\mathscr{R} = \mathscr{M}_3(\mathbb{F}_p[u]/\langle u^3\rangle) \cong \mathscr{M}_3(\mathbb{F}_p)[u]/\langle u^3\rangle.$$

Hence,

$$\begin{aligned}\mathscr{R} = (\mathbb{F}_{p^3} + e\mathbb{F}_{p^3} + e^2\mathbb{F}_{p^3}) + u(\mathbb{F}_{p^3} + e\mathbb{F}_{p^3} + e^2\mathbb{F}_{p^3})\\ +u^2(\mathbb{F}_{p^3} + e\mathbb{F}_{p^3} + e^2\mathbb{F}_{p^3}),\\ \text{where } e^3 = 1, u^3 = 0, ue = eu,\\ u^2e = eu^2, ue^2 = e^2u, u^2e^2 = e^2u^2. \qquad (1)\end{aligned}$$

The primary objective of establishing this isomorphism is to provide a framework for analyzing the algebraic properties of cyclic codes over $\mathscr{R}$, which also simplifies the theoretical investigation.

III. Main Results

A linear code $\mathscr{C}$ of length n over $\mathscr{R}$ is an $\mathscr{R}$-submodule of $\mathscr{R}^n$, and is said to be cyclic if for each codeword $c = (c_0, c_1, \ldots, c_{n-1}) \in \mathscr{C}$, its cyclic shift $(c_{n-1}, c_0, \ldots, c_{n-2}) \in \mathscr{C}$. A cyclic code of length n over $\mathscr{R}$ is also an ideal of $\mathscr{R}_n = \mathscr{R}[x]/\langle x^n - 1\rangle$ where each codeword $(c_0, c_1, \ldots, c_{n-1})$ corresponds to a polynomial $c_0 + c_1x + \cdots + c_{n-1}x^{n-1}$. We restrict ourselves to cyclic codes of length n such that $\gcd(n, p) = 1$, i.e., $p \nmid n$.

Let $\mathscr{R}[x]$ be the ring of polynomials over $\mathscr{R}$. For any $a \in \mathscr{R}$, $\overrightarrow{a}$ be the polynomial reduction of a modulo u and e as defined in Equation (II). We consider a polynomial reduction mapping $\Delta : \mathscr{R}[x] \to \mathbb{F}_{p^3}[x]$ defined by

$$\sum_{i=0}^{n-1} a_i x^i \mapsto \sum_{i=0}^{n-1} \overrightarrow{a_i} x^i.$$

Then a polynomial $f(x)$ is called basic irreducible over $\mathscr{R}$ if $\Delta(f(x))$ is irreducible over $\mathbb{F}_{p^3}$. If n is such that $p \nmid n$ and $x^n - 1 = \prod_{r=1}^{m} f_r$ where f_r's are irreducible polynomials in indeterminate x over $\mathbb{F}_{p^3}$, then the polynomial $x^n - 1$ can be uniquely factorized into pairwise coprime polynomials over $\mathbb{F}_{p^3}$.

Theorem 1 ([11], [17]). *Let n be such that $p \nmid n$ and $x^n - 1 = \prod_{r=1}^{m} f_r$ where f_r's are basic irreducible polynomials over $\mathscr{R}$. Then*

$$\mathscr{R}_n = \mathscr{R}[x]/\langle x^n - 1\rangle = \bigoplus_{r=1}^{m} \mathscr{R}[x]/\langle f_r\rangle.$$

Since $\langle f_r\rangle$ is not a two-sided ideal, we consider $\mathscr{R}[x]/\langle f_r\rangle$ as a right $\mathscr{R}$-module. Now, we find all the possible forms of the submodules of $\mathscr{R}[x]/\langle f_r\rangle$.

Theorem 2. *Let f be a basic irreducible polynomial over $\mathscr{R}$. The following is the complete set of right $\mathscr{R}$-submodules of $\mathscr{R}_f = \mathscr{R}[x]/\langle f\rangle$.*

1) $\{0\}, \langle u\rangle, \langle u^2\rangle, \langle ue\rangle, \langle ue^2\rangle, \langle u^2e\rangle, \langle u^2e^2\rangle, \langle e\rangle, \langle e^2\rangle$.
2) $\langle u + e\eta_{21} + e^2\eta_{31}\rangle$, $\langle u^2 + e\eta_{22} + e^2\eta_{32}\rangle$, $\langle u, e, e^2\rangle$ *and* $\langle u^2, e, e^2\rangle$.

where η_{mn} are units in $\mathbb{F}_{p^3}[x]$.

Proof. Let $g + \langle f\rangle$ be a nonzero element of the $\mathscr{R}$-submodule I of $\mathscr{R}_f$. Then $g \notin \langle f\rangle$ and there exist $g_{11}, g_{12}, g_{13}, g_{21}, g_{22}, g_{23}, g_{31}, g_{32}, g_{33}$ in $\mathbb{F}_{p^3}$ such that

$$\begin{aligned}g = (g_{11} + ug_{12} + u^2g_{13}) + e(g_{21} + ug_{22} + u^2g_{23})\\ +e^2(g_{31} + ug_{32} + u^2g_{33}).\end{aligned}$$

Now, we consider following two cases.

(i) If we take $\gcd(g_{11}, f) = 1$, then there exists $g_{11}^{-1} \in \mathbb{F}_{p^3}[x]$ such that $g_{11}g_{11}^{-1} + \langle f\rangle = 1 + \langle f\rangle$. Thus, $I = \mathscr{R}_f$.
(ii) If we take $\gcd(g_{11}, f) = f$, then

$$\begin{aligned}g + \langle f\rangle = (ug_{12} + u^2g_{13}) + e(g_{21} + ug_{22} + u^2g_{23})\\ +e^2(g_{31} + ug_{32} + u^2g_{33}).\end{aligned}$$

From case (ii), if we consider $\gcd(g_{21}, f) = \gcd(g_{31}, f) = f$, and $\gcd(g_{12}, f) = 1$, there exist g_{12}^{-1} such that $g_{12}g_{12}^{-1} = 1$ and we have $I = \langle u\rangle$. If along with g_{21} and g_{31}, $\gcd(g_{12}, f) = \gcd(g_{22}, f) = \gcd(g_{32}, f) = f$, and $\gcd(g_{13}, f) = 1$, we have g_{13}^{-1} such that $g_{13}g_{13}^{-1} = 1$ and $I = \langle u^2\rangle$.

If we consider $\gcd(g_{11}, f) = \gcd(g_{12}, f) = \gcd(g_{13}, f) = f$, $\gcd(g_{21}, f) = \gcd(g_{31}, f) = f$, and $\gcd(g_{22}, f) = 1$, then there exists g_{22}^{-1} such that $g_{22}g_{22}^{-1} = 1$ and we have $I = \langle eu\rangle$. Following the similar process, we can show that $I = \langle eu^2\rangle$, $I = \langle e^2u\rangle$ and $\langle e^2u^2\rangle$.

If we consider $\gcd(g_{11}, f) = \gcd(g_{12}, f) = f$, and $\gcd(g_{21}, f) = 1$, there exists g_{21}^{-1} such that $g_{21}g_{21}^{-1} = 1$ and we have $I = \langle e\rangle$. Following the similar process, we can show that $I = \langle e^2\rangle$.

Let $\gcd(g_{22}, f) = \gcd(g_{23}, f) = \gcd(g_{32}, f) = \gcd(g_{33}, f) = f$, and $\gcd(g_{21}, f) = \gcd(g_{31}, f) = 1$. Now, we have the following cases.

(I) If we consider $\gcd(g_{13}, f) = f$ and $\gcd(g_{12}, f) = 1$, we have
  (a1) $\langle u + e\eta_{21} + e^2\eta_{31}\rangle = I$
  (b1) $\langle u + e\eta_{21} + e^2\eta_{31}\rangle \subsetneq I$.
(II) If we consider $\gcd(g_{12}, f) = f$ and $\gcd(g_{13}, f) = 1$, we have
  (a2) $\langle u^2 + e\eta_{22} + e^2\eta_{32}\rangle = I$
  (b2) $\langle u^2 + e\eta_{22} + e^2\eta_{32}\rangle \subsetneq I$.

Here, $\eta_{21} = g_{21}g_{12}^{-1}$, and $\eta_{31} = g_{31}g_{12}^{-1}$, $\eta_{22} = g_{21}g_{13}^{-1}$, and $\eta_{32} = g_{31}g_{13}^{-1}$. Now, from cases (b1) and (b2) we consider the following subcases.

(i) There exist $u^i + e\eta'_{2i} + e^2\eta_{3i}$ where $\eta'_{2i} \in \mathbb{F}_{16}[x]$ such that it does not belong to $\langle u^i + e\eta_{2i} + e^2\eta_{3i}\rangle$. This implies $e(\eta_{2i} - \eta'_{2i}) \in I$. But $(\eta_{2i} - \eta'_{2i}) \notin I$. Hence $e \in I$. Hence, $I = \langle u^i, e\rangle$ for $i = 1, 2$.
(ii) There exists $u^i + e\eta_{2i} + e^2\eta'_{3i}$ where $\eta'_{3i} \in \mathbb{F}_{16}[x]$ such that it does not belong to $\langle u^i + e\eta_{2i} + e^2\eta_{3i}\rangle$. This implies $e^2(\eta_{3i} - \eta'_{3i}) \in I$. But $(\eta_{3i} - \eta'_{3i}) \notin I$. Hence $e^2 \in I$. Hence, $I = \langle u^i, e^2\rangle$ for $i = 1, 2$.

From subcase (i), and subcase (ii), $I = \langle u^i, e, e^2\rangle$ for $i = 1, 2$. □

Authorized licensed use limited to: SR University. Downloaded on November 12,2026 at 06:28:05 UTC from IEEE Xplore. Restrictions apply.

Let f_i be a factor of x^n-1 and consider $\hat{f}_i = \frac{x^n-1}{f_i}$. Then, we have the following result for the $\mathscr{R}$-modules in $\frac{\mathscr{R}[x]}{\langle x^n-1\rangle}$.

Corollary 1. *Any right $\mathscr{R}$-module in $\frac{\mathscr{R}[x]}{\langle x^n-1\rangle}$ is sum of the following $\mathscr{R}$-submodules for $x^n-1 = g_1g_2\dots g_m$, where $g_j(1 \le j \le m)$ are irreducible pairwise coprime monic polynomials in $\mathbb{F}_{p^3}[x]$.*

1) $\langle u\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle u^2\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle e\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle e^2\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle u^2e\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle u^2e^2\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle ue^2\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle ue\hat{g}_j + \langle x^n-1\rangle\rangle$.
2) $\langle (u+e\eta_{21}+e^2\eta_{31})\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle (u^2+e\eta_{22}+e^2\eta_{32})\hat{g}_j + \langle x^n-1\rangle\rangle$.
3) $(\langle u\hat{g}_j + \langle x^n-1\rangle\rangle + \langle e\hat{g}_j + \langle x^n-1\rangle\rangle + \langle e^2\hat{g}_j + \langle x^n-1\rangle\rangle)$, $(\langle u^2\hat{g}_j + \langle x^n-1\rangle\rangle + \langle e\hat{g}_j + \langle x^n-1\rangle\rangle + \langle e^2\hat{g}_j + \langle x^n-1\rangle\rangle)$.

Proof. Follows from Theorem 1 and Theorem 2. □

Throughout the paper, we use $c_0+c_1x+\dots+c_{n-1}x^{n-1}$ for the corresponding coset $c_0+c_1x+\dots+c_{n-1}x^{n-1}+\langle x^n-1\rangle$ in the ring $\frac{\mathscr{R}[x]}{\langle x^n-1\rangle}$. Now, we provide the main result of this paper.

Theorem 3. *Let $\mathscr{C}$ be a cyclic code of odd length n over $\mathscr{R}$. Then there exist units η_{ij} in $\mathbb{F}_{p^3}[x]$ and a family of irreducible pairwise coprime monic polynomials P_i, for $0 \le i \le 11$, in $\mathbb{F}_{p^3}[x]$ such that $\prod_{i=0}^{11} P_i = x^n-1$, and*

$$\mathscr{C} = \langle \hat{P_1}\rangle \oplus \langle u\hat{P_2}\rangle \oplus \langle u^2\hat{P_3}\rangle \oplus \langle e\hat{P_4}\rangle \oplus \langle ue\hat{P_5}\rangle$$
$$\oplus\langle u^2e\hat{P_6}\rangle \oplus \langle e^2\hat{P_7}\rangle \oplus \langle ue^2\hat{P_8}\rangle \oplus \langle u^2v^2\hat{P_9}\rangle$$
$$\oplus\langle (u+e\eta_{21}+e^2\eta_{31})\hat{P_{10}}\rangle \oplus \langle (u^2+e\eta_{22}+e^2\eta_{32})\hat{P_{11}}\rangle.$$

Moreover, $|\mathscr{C}| = p^{3\xi}$ where

$$\xi = 9\deg(P_1) + 6\deg(P_2) + 3\deg(P_3) + 6\deg(P_4)$$
$$+4\deg(P_5) + 2\deg(P_6) + 3\deg(P_7) + 2\deg(P_8)$$
$$+\deg(P_9) + 8\deg(P_{10}) + 7\deg(P_{11}).$$

Proof. Let $x^n-1 = f_1f_2\cdots f_m$ where $f_j(1 \le i \le m)$ are irreducible and pairwise coprime monic polynomials in $\mathbb{F}_p^3[x]$. Then from the above corollary, any right $\mathscr{R}$-module in $\frac{\mathscr{R}[x]}{\langle x^n-1\rangle}$ is the sum of the following $\mathscr{R}$-submodules.

1) $\langle u\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle u^2\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle e\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle e^2\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle u^2e\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle u^2e^2\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle ue^2\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle ue\hat{g}_j + \langle x^n-1\rangle\rangle$.
2) $\langle (u+e\eta_{21}+e^2\eta_{31})\hat{g}_j + \langle x^n-1\rangle\rangle$, $\langle (u^2+e\eta_{22}+e^2\eta_{32})\hat{g}_j + \langle x^n-1\rangle\rangle$.

Without loss of generality, we can assume that

$$\mathscr{C} = \langle \hat{f}_{m_1+1}\rangle \oplus \dots \oplus \langle \hat{f}_{m_1+\dots+m_2}\rangle \oplus \langle u\hat{f}_{m_1+m_2+1}\rangle$$
$$\oplus \dots \oplus \langle u\hat{f}_{m_1+\dots+m_3}\rangle \oplus \langle u^2\hat{f}_{m_1+m_2+m_3+1}\rangle \oplus \dots \oplus$$
$$\langle u^2\hat{f}_{m_1+\dots+m_4}\rangle \oplus \langle e\hat{f}_{m_1+\dots+m_4+1}\rangle \oplus \dots \oplus \langle e\hat{f}_{m_1+\dots+m_5}\rangle$$
$$\oplus\langle ue\hat{f}_{m_1+\dots+m_5+1}\rangle \oplus \dots \oplus \langle ue\hat{f}_{m_1+\dots+m_6}\rangle$$
$$\oplus\langle u^2e\hat{f}_{m_1+\dots+m_6+1}\rangle \oplus \dots \oplus \langle u^2e\hat{f}_{m_1+\dots+m_7}\rangle$$
$$\oplus\langle e^2\hat{f}_{m_1+\dots+m_7+1}\rangle \oplus \dots \oplus \langle e^2\hat{f}_{m_1+\dots+m_8}\rangle$$
$$\oplus\langle ue^2\hat{f}_{m_1+\dots+m_8+1}\rangle \oplus \dots \oplus \langle ue^2\hat{f}_{m_1+\dots+m_9}\rangle$$
$$\oplus\langle u^2e^2\hat{f}_{m_1+\dots+m_9+1}\rangle \oplus \dots \oplus \langle u^2e^2\hat{f}_{m_1+\dots+m_{10}}\rangle$$
$$\oplus\langle (u+e\eta_{21}+e^2\eta_{31})\hat{f}_{m_1+\dots+m_{10}+1}\rangle$$
$$\oplus \dots \oplus \langle (u+e\eta_{21}+e^2\eta_{31})\hat{f}_{m_1+\dots+m_{11}}\rangle$$
$$\oplus\langle (u^2+e\eta_{22}+e^2\eta_{32})\hat{f}_{m_1+\dots+m_{11}+1}\rangle$$
$$\oplus \dots \oplus \langle (u^2+e\eta_{22}+e^2\eta_{32})\hat{f}_m\rangle,$$

where integers $m_1, m_2, m_3, m_4, m_5, m_6, m_7, m_8, m_9, m_{10}, m_{11} \ge 0$ and $m_1+\dots+m_{11}+1 \le m$. Let $m_0 = 0$ and m_{12} be a non-negative integer such that $m_1+\dots+m_{12} = m$. Now, we define

$$P_0 = f_{m_0+1}\cdots f_{m_0+m_1}, P_1 = f_{m_0+m_1+1}\cdots f_{m_0+m_1+m_2},$$
$$P_2 = f_{m_0+m_1+m_2+1}\cdots f_{m_0+\dots+m_3},$$
$$P_3 = f_{m_0+\dots+m_3+1}\cdots f_{m_0+\dots+m_4},$$
$$P_4 = f_{m_0+\dots+m_4+1}\cdots f_{m_0+\dots+m_5},$$

and simillarly for $P_5, P_6, \dots, P_{11}$ such that

$$P_{11} = f_{m_0+\dots+m_{11}+1}\cdots f_m.$$

Then, from the above construction, $P_0, P_1, \dots, P_{12}$ are pairwise coprime and $\prod_{i=0}^{12} P_i = x^n-1$. Hence,

$$\mathscr{C} = \langle \hat{P_1}\rangle \oplus \langle u\hat{P_2}\rangle \oplus \langle u^2\hat{P_3}\rangle \oplus \langle e\hat{P_4}\rangle \oplus \langle ue\hat{P_5}\rangle$$
$$\oplus\langle u^2e\hat{P_6}\rangle \oplus \langle e^2\hat{P_7}\rangle \oplus \langle ue^2\hat{P_8}\rangle \oplus \langle u^2v^2\hat{P_9}\rangle$$
$$\oplus\langle (u+e\eta_{21}+e^2\eta_{31})\hat{P_{10}}\rangle \oplus \langle (u^2+e\eta_{22}+e^2\eta_{32})\hat{P_{11}}\rangle.$$

Furthermore,

$$|\langle \hat{P_1}\rangle| = p^{3(9\deg(P_1))}, |\langle u\hat{P_2}\rangle| = p^{3(6\deg(P_2))},$$
$$|\langle u^2\hat{P_3}\rangle| = p^{3(3\deg(P_3))}, |\langle e\hat{P_4}\rangle| = p^{3(6\deg(P_4))},$$
$$|\langle ue\hat{P_5}\rangle| = p^{3(4\deg(P_5))}, |\langle u^2e\hat{P_6}\rangle| = p^{3(2\deg(P_6))},$$
$$|\langle e^2\hat{P_7}\rangle| = p^{3(3\deg(P_7))}, |\langle ue^2\hat{P_8}\rangle| = p^{3(2\deg(P_8))},$$
$$|\langle u^2e^2\hat{P_9}\rangle| = p^{3(1\deg(P_9))},$$
$$|\langle (u+e\beta_{11}+e^2\beta_{21})\hat{P_{10}}\rangle| = p^{3(8\deg(P_{10}))},$$
$$|\langle (u^2+e\beta_{12}+e^2\beta_{22})\hat{P_{11}}\rangle| = p^{3(7\deg(P_{11}))},$$

Hence, $|\mathscr{C}| = p^{3\xi}$, where

$$\xi = 9\deg(P_1) + 6\deg(P_2) + 3\deg(P_3) + 6\deg(P_4)$$
$$+4\deg(P_5) + 2\deg(P_6) + 3\deg(P_7) + 2\deg(P_8)$$
$$+\deg(P_9) + 8\deg(P_{10}) + 7\deg(P_{11}).$$

□

Now, for an ideal $\langle x^n-1\rangle_{\mathbb{F}_{p^3}[x]}$ of $\mathbb{F}_{p^3}[x]$ generated by x^n-1, we consider $\mathscr{R}' = \frac{\mathbb{F}_{p^3}[x]}{\langle x^n-1\rangle_{\mathbb{F}_{p^3}[x]}}$. In Theorem 3, we investigated the structure of cyclic code $\mathscr{C}$ over $\mathscr{R}$.

Authorized licensed use limited to: SR University. Downloaded on November 12,2026 at 06:28:05 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CYCLIC CODES OVER $\mathbb{F}_{125}, \mathbb{F}_{27}$, AND $\mathbb{F}_8$

n	p	Generators of cyclic codes	Images under φ	Remark
3	5	$\langle (w^3u^2e + w^{13}u^2 + w^7ue^2 + w^{31}ue + 4w^{19})f_1, (w^5u^2e + w^{17}u^2w^2 + w^{23}ue^2 + 4w^{47})f_2\rangle$	$[27, 22, 5]_{125}$	AMDS
3	5	$\langle (w^3u^2e + 4w^{61}u^2 + 4w^{85})f_1, (w^5u^2e + 3u^2w^2 + 4w^34)f_2\rangle$	$[27, 25, 3]_{125}$	MDS
3	5	$\langle (w^3u^2e + w^13u^2 + w^7ue^2 + 4w^8)f_1, (w^5u^2e + w^{17}u^2w^2 + w^{23}ue^2 + 4w^{47})f_2\rangle$	$[27, 23, 4]_{125}$	AMDS
3	2	$\langle (w^3u^2e^2 + w^5u^2 + w^4ue^2 + w)f_1, (w^3u^2e + u^2w^6 + w^4ue^2)f_2\rangle$	$[27, 23, 3]_8$	Near Optimal [6]
3	2	$\langle (w^3u^2e + w^5u^2 + w^2)f_1, (w^3u^2e + u^2w^6 + w^4)f_2\rangle$	$[27, 25, 2]_8$	Optimal [6]
5	3	$\langle (w^3u^2e + w^5u^2 + w^2ue^2 + uew^2 + 2w^{18})f_1, (w^3u^2e + u^2w^6 + w^3ue^2 + uew + 2w^{11})f_2\rangle$	$[45, 39, 4]_{27}$	Near Optimal [10]
5	3	$\langle (w^3u^2e + w^5u^2 + w^2ue^2 + uew^2 + uw^{18} + 2w^5)f_1, (w^3u^2e + u^2w^6 + w^3ue^2 + uew + 2w^{11})f_2\rangle$	$[45, 38, 5]_{27}$	Near Optimal [10]
5	3	$\langle (w^3u^2e + w^5u^2 + ue^2w^24 + 2w^{11})f_1, (w^3u^2e + u^2w^6 + 2w^4)f_2\rangle$	$[45, 42, 3]_{27}$	Optimal [10]
5	2	$\langle (w^3u^2e + w^5u^2 + w^2ue^2 + uew^2 + w^2)f_1, (w^3u^2e + u^2w^6 + w^3ue^2 + uew + w^5)f_2\rangle$	$[45, 39, 4]_8$	Near Optimal [6]
5	2	$\langle (w^3u^2e + w^5u^2 + w^2)f_1, (w^3u^2e + u^2w^6 + w^3ue^2 + uew + w^5)f_2\rangle$	$[45, 41, 3]_8$	Near Optimal [6]

Now, we provide the structure of $\mathscr{C}$ over $\mathscr{R}'$.

Theorem 4. *Let $\mathscr{C}$ be a cyclic code of odd length n over $\mathscr{R}$. Then there exist polynomials $W_1, W_2, W_3, \ldots, W_{11}$ in $\mathbb{F}_{p^3}[x]$ which are factors of $x^n - 1$ such that*

$$\mathscr{C} = \langle W_1\rangle_{\mathscr{R}'} + u\langle W_2\rangle_{\mathscr{R}'} + u^2\langle W_3\rangle_{\mathscr{R}'} + e\langle W_4\rangle_{\mathscr{R}'} + ue\langle W_5\rangle_{\mathscr{R}'}$$
$$+u^2e\langle W_6\rangle_{\mathscr{R}'} + e^2\langle W_7\rangle_{\mathscr{R}'} + ue^2\langle W_8\rangle_{\mathscr{R}'} + u^2e^2\langle W_9\rangle_{\mathscr{R}'}$$
$$+(u + e\eta_{21} + e^2\eta_{31})\langle W_{10}\rangle_{\mathscr{R}'} + (u^2 + e\eta_{22} + e^2\eta_{32})\langle W_{11}\rangle_{\mathscr{R}'}$$

where η_{mn} are units in $\mathbb{F}_{p^3}[x]$ and $\langle -\rangle_{\mathscr{R}'}$ is an ideal of $\mathscr{R}'$ generated by $-$. Also,

$$|\mathscr{C}| = 16^{11n-(\deg W_1+\deg W_2+\cdots+\deg W_{11})}.$$

Proof. From Theorem 3, we have

$$\mathscr{C} = \langle \hat{P}_1\rangle \oplus \langle u\hat{P}_2\rangle \oplus \langle u^2\hat{P}_3\rangle \oplus \langle e\hat{P}_4\rangle \oplus \langle ue\hat{P}_5\rangle$$
$$\oplus\langle u^2e\hat{P}_6\rangle \oplus \langle e^2\hat{P}_7\rangle \oplus \langle ue^2\hat{P}_8\rangle \oplus \langle u^2v^2\hat{P}_9\rangle$$
$$\oplus\langle (u + e\eta_{21} + e^2\eta_{31})\hat{P}_{10}\rangle \oplus \langle (u^2 + e\eta_{22} + e^2\eta_{32})\hat{P}_{11}\rangle.$$

It is easy to show that

$$\langle \hat{P}_i\rangle = \langle \hat{P}_i\rangle_{\mathscr{R}'} \oplus u\langle \hat{P}_i\rangle_{\mathscr{R}'} \oplus u^2\langle \hat{P}_i\rangle_{\mathscr{R}'}$$
$$\oplus e\langle \hat{P}_i\rangle_{\mathscr{R}'} \oplus ue\langle \hat{P}_i\rangle_{\mathscr{R}'} \oplus u^2e\langle \hat{P}_i\rangle_{\mathscr{R}'}$$
$$\oplus e^2\langle \hat{P}_i\rangle_{\mathscr{R}'} \oplus ue^2\langle \hat{P}_i\rangle_{\mathscr{R}'} \oplus u^2e^2\langle \hat{P}_i\rangle_{\mathscr{R}'}.$$

Now, we substitute the value of $\langle \hat{P}_i\rangle$ in $\mathscr{C}$ which gives

$$\mathscr{C} = \langle \hat{P}_1\rangle_{\mathscr{R}'} + u(\langle \hat{P}_1\rangle_{\mathscr{R}'} + \langle \hat{P}_2\rangle_{\mathscr{R}'})$$
$$+u^2(\langle \hat{P}_1\rangle_{\mathscr{R}'} + \langle \hat{P}_2\rangle_{\mathscr{R}'} + \langle \hat{P}_3\rangle_{\mathscr{R}'}) + e(\langle \hat{P}_1\rangle_{\mathscr{R}'} + \langle \hat{P}_4\rangle_{\mathscr{R}'})$$
$$+ue(\langle \hat{P}_1\rangle_{\mathscr{R}'} + \langle \hat{P}_2\rangle_{\mathscr{R}'} + \langle \hat{P}_4\rangle_{\mathscr{R}'} + \langle \hat{P}_5\rangle_{\mathscr{R}'})$$
$$+u^2e(\langle \hat{P}_1\rangle_{\mathscr{R}'} + \langle \hat{P}_2\rangle_{\mathscr{R}'} + \langle \hat{P}_4\rangle_{\mathscr{R}'} + \langle \hat{P}_5\rangle_{\mathscr{R}'} + \langle \hat{P}_6\rangle_{\mathscr{R}'})$$
$$+e^2(\langle \hat{P}_1\rangle_{\mathscr{R}'} + \langle \hat{P}_7\rangle_{\mathscr{R}'})$$
$$+ue^2(\langle \hat{P}_1\rangle_{\mathscr{R}'} + \langle \hat{P}_2\rangle_{\mathscr{R}'} + \langle \hat{P}_4\rangle_{\mathscr{R}'} + \langle \hat{P}_5\rangle_{\mathscr{R}'}$$
$$+\langle \hat{P}_7\rangle_{\mathscr{R}'} + \langle \hat{P}_8\rangle_{\mathscr{R}'})$$
$$+u^2e^2(\langle \hat{P}_1\rangle_{\mathscr{R}'} + \langle \hat{P}_2\rangle_{\mathscr{R}'} + \langle \hat{P}_3\rangle_{\mathscr{R}'} + \langle \hat{P}_4\rangle_{\mathscr{R}'}$$
$$+\langle \hat{P}_5\rangle_{\mathscr{R}'} + \langle \hat{P}_6\rangle_{\mathscr{R}'} + \langle \hat{P}_7\rangle_{\mathscr{R}'} + \langle \hat{P}_8\rangle_{\mathscr{R}'} + \langle \hat{P}_9\rangle_{\mathscr{R}'})$$
$$+(u + e\eta_{21} + e^2\eta_{31})\langle \hat{P}_{10}\rangle_{\mathscr{R}'} + (u^2 + e\eta_{22} + e^2\eta_{32})\langle \hat{P}_{11}\rangle_{\mathscr{R}'}.$$

We complete the rest of the proof by proving that

$$\langle \hat{W}_2\rangle_{\mathscr{R}'} = \langle \hat{P}_1\rangle_{\mathscr{R}'} + \langle \hat{P}_2\rangle_{\mathscr{R}'},$$
$$\langle \hat{W}_3\rangle_{\mathscr{R}'} = \langle \hat{P}_1\rangle_{\mathscr{R}'} + \langle \hat{P}_2\rangle_{\mathscr{R}'} + \langle \hat{P}_3\rangle_{\mathscr{R}'},$$
$$\langle \hat{W}_4\rangle_{\mathscr{R}'} = \langle \hat{P}_1\rangle_{\mathscr{R}'} + \langle \hat{P}_4\rangle_{\mathscr{R}'}, \ \ldots\ , \langle \hat{W}_{11}\rangle_{\mathscr{R}'} = \langle \hat{P}_{11}\rangle_{\mathscr{R}'}.$$

Now, from the hypothesis we have $x^n - 1|\hat{P}_i\hat{P}_j$, and $\hat{P}_i\hat{P}_j = 0$ in $\frac{\mathscr{R}[x]}{\langle x^n-1\rangle}$ for any distinct $i, j \in \{1, \ldots, 11\}$. As $\{P_j, \hat{P}_j\}$ are pairwise coprime, there exist $y_{0j}, y_{1j} \in \mathbb{F}_{p^3}[x]$ such that $y_{0j}P_j + y_{1j}\hat{P}_j = 1$ for $j = \{1, \ldots, 11\}$. Further, if we consider $y_{0j}P_j + y_{1j}\hat{P}_j = 1$ for $j = \{2, \ldots, 11\}$, there exist polynomials $u_1, u_2, \ldots, u_{11}$ in $\mathbb{F}_{p^3}[x]$ such that

$$u_1P_2\cdots P_{11} + u_2\hat{P}_2P_3\cdots P_{11} + \cdots + u_{11}P_2P_3\cdots P_{10}\hat{P}_{11} = 1.$$

If we multiply the above equation by $\hat{P}_1$, we have $\hat{P}_1 = u_1\hat{P}_1P_2\cdots P_{11}$. From the above construction, we can conclude that $u_1W_1P_2\cdots P_{11} = \hat{P}_1$ which implies that $\langle W_1\rangle_{\mathscr{R}'} = \langle \hat{P}_1\rangle$. Now, $W_2 = \hat{P}_1 + \hat{P}_2$. Hence, $u_1W_2P_2\cdots P_{11} = \hat{P}_1$ which implies that $\hat{P}_1 \in \langle W_2\rangle_{\mathscr{R}'}$. If we continue this process, we get $\hat{P}_2 \in \langle W_2\rangle_{\mathscr{R}'}$. Hence, we can conclude that

$$\langle W_2\rangle_{\mathscr{R}'} = \langle \hat{P}_1\rangle_{\mathscr{R}'} + \langle \hat{P}_2\rangle_{\mathscr{R}'}.$$

Similarly, rest will follow. Since $|\mathbb{F}_{p^3}| = p^3$, $|\langle W_j\rangle_{\mathscr{R}'}| = p^{3(n-\deg W_j)}$ and hence

$$|\mathcal{C}| = p^{3(11n-(\sum_{j=1}^{11}\deg W_j))}.$$

□

IV. THE GRAY MAP

In the case of ring $\mathbb{F}_2 + u\mathbb{F}_2$ in [3], Lee weight is defined by $\omega_L(0) = 0, \omega_L(1) = \omega_L(1 + u) = 1, \omega_L(u) = 2$, and the Gray map from $(\mathbb{F}_2 + u\mathbb{F}_2)^n$ to $\mathbb{F}_2^{2n}$ sends $(a + ub)$ to $(b, a + b)$ for $a, b \in \mathbb{F}_2^n$. We extend this Gray map φ from $\mathscr{R}$ to $\mathbb{F}_{p^3}^9$ by

$$\varphi[(y_1 + ey_2 + e^2y_3) + u(y_4 + ey_5 + e^2y_6)$$
$$+u^2(y_7 + ey_8 + e^2y_9)] =$$

Authorized licensed use limited to: SR University. Downloaded on November 12,2026 at 06:28:05 UTC from IEEE Xplore. Restrictions apply.

$$(y_9, y_8 + y_9, y_7 + y_9, y_6 + y_9, \ldots, y_3 + y_9, y_2 + y_9,$$
$$y_1 + y_2 + y_3 + \cdots + y_9),$$

where $y_i \in \mathbb{F}_{p^3}$ for $i = 1$ to 9 and φ is also distance preserving. Hence, for any element

$$r = (y_1 + ey_2 + e^2y_3) + u(y_4 + ey_5 + e^2y_6)$$
$$+u^2(y_7 + ey_8 + e^2y_9)$$

in $\mathscr{R}$, the extended Lee weight is defined as

$$\omega_L(r) = (\omega_H(y_9) + \omega_H(y_8 + y_9) + \omega_H(y_7 + d_9) +$$
$$\omega_H(y_6 + d_9) + \cdots + \omega_H(y_3 + y_9)$$
$$+\omega_H(y_2 + y_9) + \omega_H(y_1 + y_2 + y_3 + \cdots + y_9)),$$

where $\omega_H(.)$ denotes the Hamming weight in $\mathbb{F}_{p^3}$. If we take $c = (c_0, c_1, \ldots, c_{n-1})$ in $\mathscr{R}^n$, then the Lee weight in c is $\omega_L(c) = \omega_L(c_0)+\omega_L(c_1)+\cdots+\omega_L(c_{n-1})$. The number of non-zero components is the Hamming weight of a codeword $\omega_H(c)$. Hamming distance for any two codewords c and c' is defined as $d_H(c, c') = \omega_H(c-c')$. The Hamming distance for the code $\mathscr{C}$ is defined as $d_L(\mathscr{R}) = min\{d_L(c, c') \mid c \neq c' \ \forall \ c, c' \in \mathscr{C}\}$.

Theorem 5. *For a prime $(p \leq 100)$, let $\mathscr{C}$ be a cyclic code of length n over $\mathscr{R}$, size M, and Lee distance d_L. Then $\varphi(\mathscr{C})$ is a linear code over $\mathbb{F}_{p^3}$ of length $9n$, size M, and Hamming distance d_H.*

V. Examples

A linear q-ary $[n, k, d]$ code is said to be MDS (maximum distance separable) if $d = n - k + 1$, i.e., if it meets the Singleton bound $d \leq n - k + 1$. One of the basic motives in the construction of an $[n, k]$ linear code is to optimize d and try to achieve $d_{MDS} = n-k+1$. A code is said to be AMDS (almost MDS) if its minimum distance $d = n - k$, and it is optimal if it attains the highest possible minimum distance for its given length and dimension. For $n = 3$ and 5, following is the construction strategy of cyclic codes over $\mathbb{F}_{125}, \mathbb{F}_{27}$, and $\mathbb{F}_8$, which are the Gray images of cyclic codes. We determine parameters using Magma Computation Software [4].

Example 1. *For $n = 3$ and $p = 5$, we have $x^3 - 1 = (x + 4)(x^2 + x + 1) \in \mathbb{F}_{125}[x]$. In this case, we consider $f_1 = x + 4$, and $f_2 = x^2 + x + 1$. The parameters of the codes obtained by the generators given in the Table I are $[27, 22, 5]_{125}$,$[27, 25, 3]_{125}$, and $[27, 23, 4]_{125}$.*

Example 2. *For $n = 3$ and $p = 2$, we have $x^3 - 1 = (x + 1)(x^2 + x + 1) \in \mathbb{F}_8[x]$. In this case, we consider $f_1 = x + 1$, and $f_2 = x^2 + x + 1$. The parameters of the codes obtained by the generators provided in the Table I are $[27, 23, 3]_8$, and $[27, 25, 2]_8$.*

Example 3. *For $n = 5$ and $p = 3$, we have $x^5 - 1 = (x + 2)(x^4 + x^3 + x^2 + x + 1) \in \mathbb{F}_{27}[x]$. In this case, we consider $f_1 = x+2$, and $f_2 = x^4+x^3+x^2+x+1$. The parameters of the codes obtained by the generators given in the Table I are $[45, 39, 4]_{27}$, $[45, 38, 5]_{27}$ and $[45, 42, 3]_{27}$.*

Example 4. *For $n = 5$ and $p = 2$, we have $x^5 - 1 = (x + 2)(x^4 + x^3 + x^2 + x + 1) \in \mathbb{F}_8[x]$. In this case, we consider $f_1 = x + 2$, and $f_2 = x^4 + x^3 + x^2 + x + 1$. The parameters of the codes obtained by the generators provided in Table I are $[45, 39, 4]_8$, and $[45, 41, 3]_8$.*

VI. Conclusion

In this paper, we have extended the work of [8] to the matrix ring $\mathscr{R} = \mathscr{M}_3(\mathbb{F}_p+u\mathbb{F}_p+u^2\mathbb{F}_p)$. We have deduced the structure of cyclic codes, which includes the cardinality of cyclic codes over the ring $\mathscr{R}$, the formation of ideals of the ring $\mathscr{R}$, and cyclic codes as direct sums of submodules. Additionally, we have defined the Gray map for the specified ring, which takes cyclic code over $\mathscr{R}$ to $\mathbb{F}_{p^3}$. Furthermore, we have presented a few non-trivial examples of linear codes over $\mathbb{F}_{125}$, $\mathbb{F}_{27}$, and $\mathbb{F}_8$ with good parameters that validate the results we have derived.

Acknowledgement

First and second authors are thankful to IIIT Bhopal and IIT Patna, respectively, for providing the research facilities.

References

[1] A. Alahmadi, H. Sboui, P. Solé, O. Yemen, "Cyclic codes over $M_2(\mathbb{F}_2)$." J.*Franklin Inst.* Vol. 350, no. 9, pp. 2837-2847, 2013.

[2] S. Bhowmick, S. Bagchi, R. K. Bandi, "Self-dual cyclic codes over $M_2(Z_4)$." *Discuss. Math. Gen. Algebra Appl.* Vol. 42, no. 2, pp. 349–362, 2022.

[3] A. Bonnecaze, P. Udaya, "Cyclic codes and self-dual codes over $\mathbb{F}_2 + u\mathbb{F}_2$." *IEEE Trans. Inform. Theory* Vol. 45, no. 4, pp. 1250-1255, 1999.

[4] W. Bosma, J. Cannon, "Handbook of Magma Functions." Univ. of Sydney, Sydney, 1995.

[5] D. F. Falcunit, V. P. Sison, "Cyclic codes over matrix ring $M_2(\mathbb{F}_p)$ and their isometric images over $\mathbb{F}_{p^2} + u\mathbb{F}_{p^2}$." *International Zurich Seminar on Communications(IZS)*, pp. 26-28, 2014.

[6] M. Grassl, "Code Tables: Bounds on the parameters of various types of codes" available at *http://www.codetables.de/* accessed on 20/03/2020.

[7] H. Islam, O. Prakash, D. K. Bhunia, "On the structure of cyclic codes over $M_2(\mathbb{F}_p + u\mathbb{F}_p)$." *Indian J Pure Appl. Math.* Vol. 53, no. 1, pp. 153–161, 2022.

[8] H. Islam, O. Prakash, D. K. Bhunia, "Algebraic structure of cyclic codes over $M_3(\mathbb{F}_p)$." *J. Discret. Math. Sci. Cryptogr*, Vol. 27, pp. $189 - 201$, 2024.

[9] R. Luo, P. Udaya, "Cyclic codes over $M_2(\mathbb{F}_2 + u\mathbb{F}_2)$." *Cryptogr. Commun.* Vol. 10, no. 6, pp. 1109-1117, 2018.

[10] MinT, "The online database for optimal parameters of (t, m, s)-nets, (t, s)-sequences, orthogonal arrays, linear codes, and OOAs!" available at *http://mint.sbg.ac.at/table.php?i=c* accessed on 5/9/2024.

[11] F. Oggier, P. Solé, J. C. Belfiore, "Codes over matrix rings for space-time coded modulations." *IEEE Trans. Inform. Theory* Vol. 58, no. 2, pp. $734 - 746$, 2012.

[12] J. Pal, S. Bhowmick, S. Bagchi, "Cyclic codes over $M_4(\mathbb{F}_2)$." *J. Appl. Math. Comput.* Vol. 60, no. 1-2, pp. 749-756, 2019.

[13] Pankaj, M. Pruthi, "Cyclic codes of prime power length from generalized cyclotomic classes of order 2^r." *J. Inf. Optim. Sci.* Vol. 39, no. 4, pp. 965-971, 2018.

[14] S. Patel, O. Prakash, H. Islam, "Cyclic codes over $M_4(\mathbb{F}_2 + u\mathbb{F}_2)$." *Cryptography Commun.* Vol. 14, no. 5, pp. 1021–1034, 2022.

[15] O. Prakash, H. Islam, S. Das, "On cyclic codes with minimal generating sets over the ring $\mathbb{Z}_{p^k}$." *Southeast Asian Bull. Math.* Vol. 45, no. 4, pp. 521-532, 2021.

[16] V. Pless, Z. Qian, "Cyclic codes and quadratic residue codes over $\mathbb{Z}_4$." *IEEE Trans. Inf. Theory* Vol. 42, no. 5, pp. 1594-1600, 1996.

[17] R. Wisbauer, "Foundations of module and ring theory." Gordon and Breach Science Publishers, 1991.

[18] J. Wolfmann, "Negacyclic and cyclic codes over $\mathbb{Z}_4$." *IEEE Trans. Inf. Theory* Vol. 45, no. 7, pp. 2527-2532, 1999.

Authorized licensed use limited to: SR University. Downloaded on November 12,2026 at 06:28:05 UTC from IEEE Xplore. Restrictions apply.